

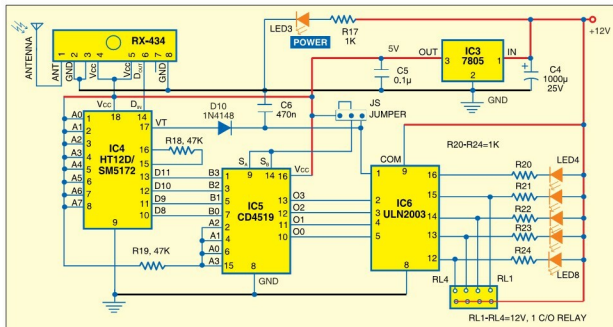


Fig. 5: Receiver circuit

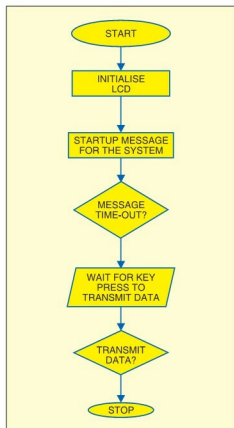


Fig. 6: Software flow-chart

respective input switches.

Data inputs AD8 through AD11 (pins 10 through 13) of HT12E are connected to the microcontroller. Pins 1 through 8 (A0 through A7) of the IC are address inputs. Shorting the address pins using switches to either Vcc or Gnd enables different address

selections for data transmission. Here we have connected them to 5V. Since address pins are connected to 5V, the address is set to 255d (in decimal). If you were to connect all the address pins to ground, the address would be 000d. Thus there are 256 possible addresses available. So you can set up switches to control one or more of the encoder address pins.

Pin 14 is a transmit-enable (TE) input pin. The encoder will send data only when pin 14 is connected to ground. Whenever a button is pressed, logic '0' is sent to this pin through the microcontroller, thus activating it and enabling transmission.

Pin 17 is the data-out (D_{out}) pin that sends a serial stream of pulses containing the address and data. It is connected to the data input pin of the TRX RF module.

The time-out control is set using input keys S1 through S4 to turn on/off the devices at predetermined time. The default time for all the devices is '00' seconds. So using 'up' key you can increment time by one second, and using 'down' key you can decrement time by one second down. At the same time, the LCD module shows the current status of increments and decrements.

When the time-out for a device is set, press 'ent' key so that the program

control transfers to the next device for time-out settings. In the same way, the remaining three time-out settings must be done before pressing 'run' key. When 'run' key is pressed, it executes the device control program subroutine in the microcontroller and the program automatically collects the time-out information entered by the user and sends the processed data to encoder IC HT12E. The encoder IC sends the data to D_{in} (pin 2) of the RF transmitter module. The data is transmitted by the transmitter from the antenna.

Receiver circuit. Fig. 5 shows the receiver circuit. The RF receiver (RX-434) module can receive the signal transmitted by the transmitter from a distance of up to 9 metres (30 feet). The range can be increased up to 30 metres using a good antenna.

D_{out} pin of RX-434 RF module is connected to D_{in} pin of decoder IC HT12D (IC4). D_{in} pin of IC4 receives address and data bits serially from the RF module. Decoder IC4 separates data and address from the received information. It accepts data only if the received address matches with the address assigned to encoder IC1 (HT12E). We have used '1111' as the permanent address for communication. Pins 1 through 8 of IC4 are address pins and therefore 256 possible addresses are available. The address on the encoder and decoder ICs must match for the data to be valid.

The HT12D decoder receives serial addresses and data from the encoder that are transmitted by a carrier signal over RF medium. The decoder compares the serial input data three times